MILITARY SPECIFICATION

SHRINKING AND STRETCHING MACHINE, SHEET METAL, MANUAL AND POWER OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manual and power operated shrinking and stretching machines for forming sheet metal, angles, flatbars, and plate with equipment and accessories specified herein.

1.2 Classification. The shrinking and stretching machines shall be of the following types and of the sizes listed in tables I, II, III, and IV. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Manually operated.
Type II - Pneumatically operated.
Type III - Hydraulically operated.
Type IV - Mechanically operated.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3441
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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, PART 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with spec fic acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

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AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI B11.7 - Safety Requirements for the Construction, Care, and Use of Cold Headers and Cold Formers.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

NATIONAL FLUID POWER ASSOCIATION INC (NFPA)

NFPA/T2.25.1M - Pneumatic Fluid Power-Systems Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fluid Power Association, Inc., 3333 N. Mayfair Road, Milwaukee WI 53222-3291.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The shrinking and stretching machine shall be new (not a prototype) and one of the manufacturer's current production models. The machine shall be capable of shrinking and stretching ferrous and non-ferrous metal through the application of compression or tension force on the metal which is gripped in sliding jaws, in accordance with the requirements herein. The machine shall consist basically of a "C" type frame with a ram for application of pressure to the

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jaws. Design of the machine shall allow the operator to change tooling from one operation to another in not more than 10 minutes. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine, When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S or the SI System of Units shall be used to graduate measuring and indicating devices such as scales, depth stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 6.2.1), measuring and indicating devices shall be graduated to read in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the intended use, performance, and design life of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station. Types II, 111, and IV machines shall have remote foot controls.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all Parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OHSA 29 CFR, Part 1910 and ANSI B11.7 that are applicable to the machine itself. If a conflict arises between OSHA 29 CFR 1910 and B11.7, 29 CFR 1910 shall apply. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market, "If

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specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workplaces. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts, either US or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28, and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), applicable equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class-and the emission and susceptibility requirement shall be as specified.

3.4 Components. The machine shall consist of not less than a frame, base, ram, and jaw assemblies. Any additional components necessary to the operation of the machine to its capacity as specified in tables I, 11, III, and IV shall be furnished.

TABLE I- Type I-manually operated, sizes and capacities.

Size	Throat depth (inches)	Capacity angle		Maximum radius required for 90° bend
		Leg width (inches)	Thickness (gauge/inches)	
1	3	1 x 1	16 ga.	3 in.
2	6	1-1/2 x 1-1/2	14 ga.	4-1/2 in.
3	6	1-1/2 x 1-1/2	0.187 in.	6 in.

TABLE II- Type II-pneumatically operated, sizes and capacities.

Size	Throat depth (inches)	Capacity angle		Maximum radius required for 90° bend
		Leg width (inches)	Thickness (gauge/inches)	
1	6	1-1/2 x 1-1/2	.250 in.	8 in.
2	12	3/4 x 3/4	.109 in.	5 in.

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TABLE III- Type III-hydraulically operated, sizes and capacities.

Size	Throat depth (inches)	Capacity angle		Maximum radius required for 90° bend
		Leg width (inches)	Thickness (gauge/inches)	
1	4	1 x 1	16 ga.	3 in.
2	4	2 x 2	1/4 in.	8 in.

TABLE IV Type IV-mechanically operated, sizes and capacities.

Size	Throat depth (inches)	Capacity angle		Maximum radius required for 90° bend
		Leg width (inches)	Thickness (gauge/inches)	
1	12	1-1/2 x 1-1/2	14 ga.	4-1/2 in.
2	18	2 x 2	11 ga.	6 in.
3	26	3 x 3	3 ga.	9 in.

NOTE : Requirements are not less than those shown in tables I, II, III, and IV.

3.4.1 Frame. The frame shall be either a steel casting or steel weldment manufactured sufficient in strength and rigidity to support and maintain alignment of all components when the machine is operated at its maximum rated capacity.

3.4.2 Base. A base shall be provided with sufficient rigidity to support the frame and all components. Mounting holes shall be provided to secure and level the base to the floor.

3.4.3 Ram or pressure lever. The machine shall have a ram or pressure lever for mounting the upper jaw assembly. The ram or pressure lever shall ‘be supported by guides or bushings mounted-on the frame, permitting vertical operation through a lever and toggle linkage.

3.4.4 Type I machines (manual). The type I machine shall be manually operated by either a foot treadl e or hand lever providing sufficient force to accomplish the shrinking and stretching capacities for the machine size specified. The foot treadle or hand lever shall be either counter balanced or spring loaded to maintain the starting position after release.

3.4.5 Type II machines (pneumatic). The type II machine shall be pneumatically operated. The pneumatic system shall conform to NFPA/T2.25.1M. The system shall have an air line filter, water trap, and oil mist lubricator, as

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necessary. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.4.6 Type III machines (hydraulic). The type III machine shall be hydraulically operated. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have a means for determining fluid level, a provision for draining, and a cleanout access plate if the reservior is not removable.

3.4.7 Type IV machines (mechanical). The type IV machine shall be electro-mechanically operated. Power shall be transmitted mechanically by V-belts from an electric motor, to a flywheel, and ram actuating mechanism, with sufficient force to perform the shrinking and stretching operations, as specified herein.

3.4.8 Jaw assemblies. The jaw assembly shall consist of an anvil and moveable jaws made of alloy steel with a surface hardness of not less than 55 on the Rockwell "C" scale. The jaw assemblies shall be mounted one to the ram or pressure fingers, and one to the bed by means of a quick-change locking and releasing device to expedite tooling changeover. The jaw assembly shall perform one forming operation per stroke. Unless otherwise specified (see 6.2.1), the standard shrinking assemblies shall have a number 2 stipple or serrated gripping surface. When surface marking is unacceptable, jaws made of plastic or other suitable material shall be provided.

3.4.9 Electrical system. The electrical system required on type II, III, and IV machines, shall conform to NFPA 79. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine, when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.9.1 Primary input voltage. Unless otherwise specified (see 6.2.1), the powered machines shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phrase 60 Hz circuit. The machine shall be initially wired to operate from a 460 volt source.

3.4.9.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

3.4.9.3 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.9.4 Control circuit voltage. The control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

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3.4.10 Hour meter. Unless otherwise specified (see 6.2.1) type II, III, and IV machines shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonsetting type and shall have a range of O to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in tables I, II, III, and IV for the machine size specified.

3.6 Performance. The machine shall be capable of shrinking or stretching metal sheets, angles, flat bars, and plates by increasing or decreasing the length of the metal in a localized area. The machine shall have the ability to remove wrinkles and produce double curvatures from flat sheets. All operations shall be performed within the capacities of tables I, II, III, and IV.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. All equipment normally furnished with the manufacturer's standard commercial machine shall be furnished, including the following:

a. One set of maintenance and adjusting tools.

b. One pair of standard stretching jaw assemblies.

c. One pair of extended shrinking jaw assemblies.

d. One pair of extended stretching jaw assemblies.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication plate. A corrosion resistant lubrication plate shall be securely attached to each machine. The plate shall contain the following information:

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Points of lubricant application
Servicing interval
Type of lubricant
viscosity

3.10.2 Name late. A corrosion resistant nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and accessories shall meet all requirements specified herein and shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 uality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 0 minutes, or a minimum of 20 strokes on manually operated machines. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The machine shall be tested for conformance with the requirements of 3.6, by shrinking and stretching two test pieces of angle iron, of not less than 60,000 psi tensile strength, and of the sizes and capacities shown in applicable tables I, II, III, or IV, for the machine size being tested. The workplaces shall be shrunk or stretched to form a 90 degree angle within the radius shown in the applicable table for the workpiece thickness being formed. The finished test pieces shall be free of stretch cracks and wrinkles.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.8, using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When

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specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The shrinking and stretching machines covered by this specification are intended for use by aircraft manufacturers, fabricators, and the sheet metal trade, to shrink and stretch flats, angles, plates, and extrusions for forming, fitting, and removing wrinkles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US or SI), if required (see 3.2.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Jaw assemblies, if different (see 3.4.8).

k. Primary input voltage, if different (see 3.4.9.1).

1. Motor enclosure, if different (see 3.4.9.2).

m. Hour meter, if different (see 3.4.10).

n. Size and capacity, if different (see 3.5).

o. Additional performance features, if required (see 3.6.1).

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p. Specify and fully describe optional equipment, if required (see 3.8).

q. Specify applicable repair parts, if required (see 3.9).

r. First article inspection, if different (see 4.4).

s. Quality conformance inspection, if different (see 4.5).

t. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification are as follows:

MIL-S-45860D	MIL-S-45860D
Type I	Type I
Type II	Type II
Size 3	Deleted
Type III	Type III
	Size 2 (new size)
	Type IV (new type)
	1 (new size)
	Size 2 (new size)
	Size 3 (new size)

Machine capacities changed from flat stock in MIL-S-45860D to angle in MIL-S-45860D.

6.5 Subject term (key word) listing.

Doming
Sheet, metal
Straightening
Stretching
Shrinking

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Preparing activity:
DLA-IP

(Project 3441-0127)

Review activities:
Air Force - 84
DLA-GS

User activities:
Army - SM
Navy - MC

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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DEFENSE LOGISTICS AGENCY
Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSG
Memphis, TN 38114-5051

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PENALTY FOR PRIVATE USE $300

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-45860D	Shrinking and Stretching Machine, Sheet Metal, Manual and Power Operated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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